1^{st} Quiz Chapter 01-Chapter 02

Name:________ Student ID:________ Instructor:________ Score:________

Part A multiple choice (select the one that is best in each case. 1point/question)

1-5 EDBDD

6-10 DEBDE

11-15 EDAAC

16-18 BEC

Part B: Short Answer (Write legibly and show all work for all steps in the problem) (67 points)

1. Rubidium has two naturally occurring isotopes, rubidium-85 (atomic mass = 84.9118 amu; abundance = 72.15%) and rubidium-87 (atomic mass = 86.9092 amu; abundance = 27.85%). (a)What is the definition of amu? (b) Calculate the atomic weight of rubidium. (4 points)

(a)1/12 mass of one ^{12}C atom

(b)the atomic weight of rubidium

= 84.9118 amu × 72.15% + 86.9092 amu × 27.85% = 85.4681 amu

So the atomic weight of rubidium is 85.4681 amu.

2. The average atomic weight of copper, which has two naturally occurring isotopes, is 63.5. One of the isotopes has an atomic weight of 62.9 amu and constitutes 69.1% of the copper isotopes. Calculate the atomic weight (amu) of the second isotope. (4 points)

The second isotope constitutes 1 – 69.1% = 30.9%

Let the second isotope atomic weight is x, then

62.9 × 69.1% + x × 30.9% = 63.5

x = 64.8

3. How many μg are there in 1 mg? (1 points)

1 g = 1000 mg = 10^6 μg

1 mg = 1000 μg

4. What is the name of the unit that equals (a) 10^{-9} gram, (b) 10^{-6} second, (c) 10^{-3} meter?(3 points)

(a)1 ng = 10^{-9} g

(b)1 μs = 10^{-6} s

(c)1 mm = 10^{-3} m

5. A weather forecaster predicts the temperature will reach 31 °C. What is this temperature (a) in K, (b) in °F? (2 points)

(a)K = °C + 273.15 = 31 + 273.15 = 304.15 K

(b)°F = 9/5(°C) + 32 = 87.8 °F

6. How many significant figures are in each of the following numbers (assume that each number is a measured quantity)? (a) 4.003, (b) $6.023*10^{23}$, (c) 5000. (3 points)

(a)4

(b)4

(c)1

7. Complete the following table. (5 points)

Symbol	$^{40}Ca^{2+}$	$^{37}Cl^{-}$
Number of protons	20	17
Number of neutrons	20	20
Number of electrons	18	18
Atomic number	20	17
Mass number	40	37
Net charge	2+	1–

8. Identify the following as a molecular or ionic compound and provide the name of chemical formula. (4 points)

(1) $AsCl_3$

molecular compound, arsenic trichloride

(2) NH_4ClO_4

ionic compound, ammonium perchlorate

(3) BaH

ionic compound, barium hydride

(4) $Mn_3(PO_4)_2$

ionic compound, manganese(II) phosphate or manganous phosphate

9. How many protons, neutrons, and electrons are in an atom of (a) ^{197}Au, (b) strontium-90? (2 points)

(a)proton: 79, electron: 79, neutron: 118

(b)proton: 38, electron: 38, neutron: 52

10. Magnesium has three isotopes with mass numbers 24, 25, and 26. (a) Write the complete chemical symbol (superscript and subscript) for each. (b) How many neutrons are in an atom of each isotope? (2 points)

(a) $^{24}_{12}Mg$, $^{25}_{12}Mg$, $^{26}_{12}Mg$

(b) $^{24}_{12}Mg$, 12 neutrons; $^{25}_{12}Mg$, 13 neutrons; $^{26}_{12}Mg$, 14 neutrons

11. Naturally occurring chlorine is 75.78% ^{35}Cl (atomic mass 34.969 amu) and 24.22% ^{37}Cl (atomic mass 36.966 amu). Calculate the atomic weight of chlorine. (2 points)

the atomic weight of chlorine

$= 34.969 \text{ amu} \times 75.78\% + 36.966 \text{ amu} \times 24.22\% = 35.453 \text{ amu}$

12. Which two of these elements would you expect to show the greatest similarity in chemical and physical properties: B, Ca, F, He, Mg, P? (2 points)

Mg and Ca

13. In which of the following species is the number of protons less than the number of electrons? (a) Ti^{2+}, (b) P^{3-}, (c) Mn, (d) Se_4^{2-}, (e) Ce^{4+}. (2 points)

(b)(d)

14. How many protons, neutrons, and electrons does the $^{79}Se^{2-}$ ion possess? (1 points)

protons: 34, electrons: 36, neutrons: 45

15. Which of these compounds would you expect to be ionic: N_2O, Na_2O, $CaCl_2$, SF_4?(2 points)

Na_2O, $CaCl_2$

16. Which of these compounds are molecular: CBr_4, FeS, P_4O_6, PbF_2? (2 points)

CBr_4, P_4O_6

17. Why is CrO named using a Roman numeral, chromium(II) oxide, whereas CaO named without a Roman numeral, calcium oxide? (2 points)

Chromium exhibits differing charges in its compounds, so we need to identify what charge it has in this case. In contrast, calcium is always 2+ in its compounds.

18. Based on the formula for the sulfate ion, predict the formula for (a) the selenate ion and (b) the selenite ion. (2 points)

(a)SeO_4^{2-}

(b)SeO_3^{2-}

19. Name the ionic compounds (a) K_2SO_4, (b) $Ba(OH)_2$, (c) $FeCl_3$. (3points)

(a) potassium sulfate

(b) barium hydroxide

(c) iron(III) chloride or ferric chloride

20. Name the compounds (a) SO_2, (b) PCl_5, (c) Cl_2O_3. (3points)

(a) sulfur dioxide

(b) phosphorus pentachloride

(c) dichlorine trioxide

21. Write the name of the following cations or anions. (16 points)

Co^{2+} cobalt(II) ion or cobaltous ion

Ca^{2+} calcium ion

Cu^{2+} copper(II) ion or cupric ion

Fe^{2+} iron(II) ion or ferrous ion

Ba^{2+} barium ion

Mn^{2+} manganese(II) ion or manganous ion

Zn^{2+} zinc ion

Hg^{2+} mercury(II) ion or mercuric ion

Pb^{2+} lead(II) ion or plumbous ion

H^- hydride ion

F^- fluoride ion

Cl^- chloride ion

Br^- bromide ion

I^- iodide ion

CN^- cyanide ion

OH^- hydroxide ion